

menler

Your turning point in the AI era.

ENGINEERING OVERVIEW

Internal Engineering Copilot

An in-IDE copilot grounded in your own repos and docs that scaffolds code, reviews it against internal context, and drafts the design notes.

Claude CLI

Cursor

Python + RAG

~40%

design-to-first-PR time reduction

4

in-IDE capabilities

Zero

manual context-wiring before first PR

One

retrieval layer over private codebase

THE PROBLEM

Generic AI doesn't know your codebase

Engineers spend the slow early hours of every task wiring up context, conventions and design before they can write a single line that actually ships.

Private Repos

Codebase the generic model has never seen

Internal Docs

Patterns and conventions not in any public dataset

Code Conventions

Naming, structure and style that exists only internally

Design History

Prior decisions locked in old PRs and Confluence

Hours spent loading context
before useful AI output is possible

No repo

awareness

Generic

not grounded

Wrong

patterns suggested

"I spend the first two hours of every new task just getting the AI to understand what we're building and how we build it. Then I can actually start."

— Composite senior engineer feedback

HOW IT WORKS

Four steps, from cold repo to shipped PR

01

Index

Repos, docs and convention files are chunked, embedded and indexed into a local RAG retrieval layer.

Repos

Docs

Conventions

Embeddings

02

Scaffold

New code is scaffolded following the team's existing patterns, pulled directly from the indexed codebase.

Pattern match

House style

Boilerplate

03

Review

Each change is reviewed against internal standards. Deviations from house conventions are flagged inline.

RAG retrieval

Convention check

Inline flag

04

Draft

A design note is drafted from the change context without leaving the editor — ready to paste into a doc or PR.

Design note

PR description

Context snippet

ARCHITECTURE

Claude through the CLI and Cursor, with a Python retrieval layer over the internal codebase

KNOWLEDGE SOURCES (RAG layer)

Private repos

Internal docs

Convention files

PR history

INTERFACE (Claude CLI + Cursor)

Claude CLI

Cursor extension

IDE commands

Inline output

CAPABILITIES (Grounded Claude)

Code scaffolding

RAG review

Convention check

Design draft

DELIVERY

Scaffolded code · Grounded review · Convention flags · Design note · All in the IDE

KEY FEATURES

Four capabilities that earn their place

No context-loading before useful work starts. Every feature serves the engineer in the IDE.

1

RAG over Private Repos and Docs

Private repos, docs and convention files indexed locally. Retrieval that knows your codebase — not the public internet.

Repos

Docs

Conventions

Embeddings

2

Pattern-Aware Code Scaffolding

New code is generated following the actual patterns in your codebase — names, structure and style that match the team.

House patterns

Style-matched

3

Grounded Code Review

Reviews run against internal standards, not generic best practices. Convention deviations are flagged with context inline.

Internal standards

Inline flags

4

Design Notes in the IDE

Draft design notes are generated from the change context without leaving the editor — ready to copy into a doc or PR.

IDE-native

PR-ready

RESULTS & IMPACT

Three numbers that justify the build

~40%

design-to-first-PR time reduction

House

conventions matched by generated code

Less

context-loading before real work starts

What this unlocks

- Engineers start building from a grounded scaffold, not from zero
- Reviews catch convention drift before it reaches the PR
- Design notes drafted in the IDE without switching tools
- Onboarding time drops — the copilot carries the codebase context
- Retrieval layer improves as the codebase grows and is re-indexed

"I used to spend the first morning on a new task just reading old PRs to understand our patterns. The copilot removed that entirely."

— Composite senior engineer feedback

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Your turning point in the AI era.

Your codebase. Your conventions. In the IDE.

The Internal Engineering Copilot grounds Claude in your own repos and docs — scaffolding code, reviewing it and drafting design notes without leaving the editor.

Thank you.